intersection of Explainability & Trustworthiness and Logic & Reasoning had 3.33% (n=9). The intersection of any of the three categories ranged from 5 to 9 entries, except for the intersection of Explainability and Trustworthiness & Logic and Reasoning & Knowledge Representation, which had none. There was only one entry at the intersection of all 4 of the main research focal areas (excluding Meta-Cognition) which was AlphaGeometry from Google[29].

4. Discussion & Open Questions

To build upon the existing literature that has thoroughly summarized the Neuro-Symbolic landscape before 2020, we extend this discussion by analyzing the most influential projects in each sub-field of Neuro-Symbolic AI published since 2020. This section aims to highlight state-of-the-art (SOTA) technologies available to researchers, showcasing significant advancements and ongoing challenges.

4.1. Knowledge Representation

Research in Knowledge Representation has focused on advancing semantic grounding, representing complex relationships, and improving data efficacy. Development of commonsense knowledge bases and event-based representations [107, 105, 106] has advanced AI's understanding of daily events, aiming to reduce error rates in text generation. These works are furthered through the exploration of minimal data requirements for commonsense knowledge in few-shot learning models[110], and the use of Neuro-Symbolic representations to enhance training efficiency and reduce costs [161]. Additionally, refinement of knowledge representation was demonstrated by predicting complex relationships and embedding techniques in knowledge graphs [111, 109], and the integration of personalized knowledge was demonstrated to ensure narrative consistency in storytelling agents [130]. NeuroQL, a domain-specific language for inter-subjective reasoning, captured complex and long-range relationships, demonstrating how Neuro-Symbolic approaches can 'do more with less', yielding significant savings in training time and environmental impact [108]. Open research questions remain around how Neuro-Symbolic AI can enhance the dynamic interpretation and manipulation of symbols, develop meta-cognitive abilities to monitor and adjust reasoning processes, and ensure transparent, explainable reasoning pathways for more human-like, adaptable, and robust knowledge representation.

4.2. Learning and Inference

Within Learning and Inference, research has focused on Neuro-Symbolic Integration for Enhanced Learning, Advanced Problem Solving and Decision Making, and Semantic Enhancement for Model Trustworthiness. Neuro-Symbolic integration for Enhanced Learning was demonstrated through the fusion of symbolic reasoning with neural learning mechanisms which adapted commonsense knowledge for few-shot settings and transformed observations into logical facts using Logical Neural Networks [84, 64]. Advanced Problem Solving and Decision Making are highlighted by Plan-SOFAI [92] and the ZeroC architecture [160], which leverage Neuro-Symbolic methods to enhance AI planning and zero-shot concept recognition to integrate the fast and slow thinking models and the improve machine generalization. Semantic Enhancement and Model Trustworthiness were demonstrated by the introduction of a Pseudo-Semantic Loss for autoregressive models which integrated logic within the loss function [122] and neural networks utilising Logic Tensor Networks [79] which aim to boost logical consistency, reduce model toxicity, and enhance prediction accuracy by concentrating on relevant constraints. Open research questions remain in Neuro-Symbolic AI, including how to develop incremental learning that allows symbolic systems to evolve with new experiences, create context-aware inference mechanisms that adjust reasoning based on situational cues, achieve fine-grained explainability for complex inference chains, and explore meta-cognitive abilities enabling systems to monitor, evaluate, and optimize their learning processes in dynamic environments.

4.3. Explainability and Trustworthiness

Research centred on Explainability and Trustworthiness within Neuro-Symbolic AI has looked to advance Natural Language Processing (NLP) Techniques, Enhancing Logical Reasoning, and Refining Language Understanding and Summarization. Braid introduced a logical reasoner with probabilistic rules to tackle the brittle matching problem, merging symbolic and neural knowledge to enhance logical reasoning [119]. Similarly, Structure-Aware Abstractive Conversation improved summarization by incorporating discourse relations, action triples, and structured graphs for precise, context-rich summaries, advancing NLP techniques [117]. Semantic-level revisions identifying and correcting "confounders" in Neuro-Symbolic scenes have improved AI decision-making clarity, fostering trust and enhancing logical reasoning by making processes more understandable [166]. Evaluating AI's humour comprehension with the New Yorker Cartoon Caption Contest underscored the need for nuanced understanding, refining language processing in complex cognitive tasks [118]. FactPEGASUS focuses on ensuring factuality in summarization by optimizing pre-training and fine-tuning methods, crucial for maintaining summary integrity and refining language understanding [135]. Complementing this, Neuro-Symbolic methods enhance explainable short answer grading through logical reasoning and cue detection, bridging AI capabilities with human-like responses and advancing NLP techniques [121]. Open research questions remain around how Neuro-Symbolic AI can adapt and evolve symbolic representations in real-time to maintain transparency, integrate meta-cognitive mechanisms for self-monitoring and adjustment of reasoning strategies, develop explainable NLP techniques for complex cognitive tasks, and ensure factual consistency in AI outputs while providing clear, detailed explanations of the underlying reasoning process.

4.4. Logic and Reasoning

From the research field of Neuro-Symbolic Logic and Reasoning, the research has gravitated largely toward the Integration of Logical Reasoning and Probabilistic Models, Commonsense Knowledge and Language Understanding, and Enhanced Decision-Making. Logical Credal Networks [36] combines logical reasoning with probabilistic models to handle imprecise information, while DeepStochLog [75] enhances traditional logic programming with neural networks for complex reasoning tasks. 2P-Kt [37] offers a comprehensive logic-based framework supporting various reasoning tasks and integrating symbolic and sub-symbolic AI. Research into

Commonsense Knowledge and Language Understanding includes kogito, [106], which generates commonsense knowledge inferences from textual input to enhance AI adaptability, and LinkBERT [88], which improves language understanding and reasoning capabilities by incorporating document links, particularly in multi-hop reasoning tasks. For Enhanced Decision-Making, "Neuro-Symbolic Commonsense Social Reasoning" [44] integrates Neuro-Symbolic methods in autonomous systems, while LASER [135] combines neural networks' flexibility with the precision of symbolic logic. "Getting from Generative AI to Trustworthy AI" [10] addresses LLMs' limitations in trustworthiness and reasoning, proposing integration with symbolic AI systems for reliability. Open research questions remain around how Neuro-Symbolic AI can develop scalable frameworks that integrate traditional logic programming with neural networks for complex reasoning tasks, incorporate commonsense knowledge and advanced language understanding to enhance multi-hop reasoning capabilities, combine symbolic logic with neural networks to ensure reliable and trustworthy decision-making and integrate meta-cognitive abilities to enable self-monitoring and adjustment of reasoning strategies for clearer, more understandable explanations.

4.5. Intersections of the above four research areas

Much of the literature is cross-sectional between the four areas of research; Explainability & Trustworthiness, Knowledge Representation, Learning & Inference and Logic & Reasoning. AlphaGeometry [29], which is a Neuro-Symbolic system designed to solve Euclidean plane geometry problems at the Olympiad level, stands out as a prominent project that sits at the intersection of all four. AlphaGeometry's ability to synthesise millions of theorems and proofs, using a neural language model trained on large-scale synthetic data to guide a symbolic deduction engine, makes it a groundbreaking example of how Neuro-Symbolic AI can achieve advanced problem-solving capabilities, bridging gaps across multiple domains of AI research. However, there is a distinct lack of integration with the explainability and trustworthiness fields within the unions of the other three research areas as the density of research intersection at the unions of explainability and trustworthiness and the other three research areas is relatively sparse, indicating a significant opportunity for further interdisciplinary work in Neuro-Symbolic AI.

4.6. Meta-Cognition

Recent advancements in this domain showcase Reinforcement learning (RL) to approximate Meta-Cognition, approaches to integrate cognitive architectures with LLMs to approximate meta-cognitive capabilities and the integration of many AI architectures and systems to demonstrate the Common Model of Cognition (CMC). The benefits of integrating symbolic features with RL algorithms were demonstrated through meta-reinforcement learning combined with logical program induction to improve financial trading strategies [173]. Enhancing general intelligence by fusing cognitive architectures with LLMs was investigated, creating embodied agents that leverage the strengths of both approaches [169, 170]. Improvements in experiential models were achieved by using LLMs to convert descriptive information into dense signals for instance-based learning [171]. Adaptive conflict resolution in AI was enhanced by coupling cognitive reasoning with generative algorithms [172]. Robust AI systems were developed through modular, agency,